

Parth Thakkar

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ABOUT

Computer Science graduate from BITS Pilani (**#1 private engineering institute in India**), currently working as a Quantitative Analyst at **Goldman Sachs** building optimization systems and LLM agents for a \$900B+ balance sheet. **US Green Card** holder (no visa sponsorship required); seeking **Software Engineering** roles in the US.

EXPERIENCE

Goldman Sachs — *Quantitative Analyst*

Jan 2024 - Present

Liability Optimizer

- Built the Python optimization engine behind **\$600B+ of firm funding** whose output drives multi-year liquidity buffer decisions and the plans reported to the **Federal Reserve**
- Cut solver runtime by 90%** for similar runs via warm starts, state reuse, and selective recomputation, allowing more iterations per cycle
- Reduced business analysis time from hours to minutes by building a run-diff tool that highlights input changes

AI Funding Planning Assistant

- Led development of a conversational **LLM agent** for the weekly funding planning cycle that surfaces binding metrics, their drivers, and week-over-week changes, **cutting time by ~60%** to create a plan
- Designed a **ReAct agent** (LangGraph + Pydantic) with a **knowledge base** and **20+ tools** (fetchers, deterministic calcs, external APIs) that can make edits to the plan with a human-in-the-loop commit path

Infrastructure & DevOps

- Built and deployed 4+ microservices on **Kubernetes** clusters with CI/CD pipelines, logging, and SSO auth
- Reduced average response time by 74% by building a **Redis** backed state cache for repeated calls
- Wrote end-to-end regression and integration test suites; drove version-controlled code reviews within CI/CD

MapMyIndia — *Developer Intern*

May 2022 - Jul 2022

- Developed a FastAPI backend to compute optimal EV routes using **Dijkstra's** algorithm on a geospatial graph
- Collaborated in a 4-member team to deliver a **full-stack** EV navigation prototype with React and MongoDB

TECHNICAL SKILLS

Programming & Tools: Python, Java, C/C++, SQL, FastAPI, REST APIs, microservices, Snowflake, Pandas, NumPy, Pydantic, Git

DevOps & Infrastructure: Kubernetes, ArgoCD, Docker, GitLab CI/CD, SonarQube, Linux

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Object-Oriented Programming, Logic in Computer Science, Discrete Math, Probability & Statistics

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani

Pilani, India

B.E.(Hons) in Computer Science — CGPA 8.5/10

Aug. 2020 – Jul. 2024

PROJECTS

Volunteer-Support Fog Computing | *Java, Networking*

2022

- Built a Java fog system across ~20 nodes with new volunteer-node **load distribution** for delay-sensitive IoT
- Achieved **31% delay reduction**, **79% energy reduction**, and **65% network usage reduction** over pure cloud

LEADERSHIP & AWARDS

Intern Coding Challenge | *Goldman Sachs*

2023

- Third place** out of **400+ participants** across India in the Intern Coding Challenge

Department of Controls | *BITS Pilani*

2020 - 2024

- Scheduling Head of the **largest student-managed sports fest** (BOSM) in India witnessing a footfall of over **10,000 students**

Microsoft Learn Student Ambassadors | *Competitive Coding SIG, BITS Pilani*

2022 - 2023

- Created online scavenger hunt problems and quizzes for 200+ participants for the annual technical festival